STACKS, DERIVED CATEGORIES OF SPACES: DETECTING-BOUNDED-BELOW

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7421-7445 of sources/stacks-spaces-perfect/pinned-original-source.tex; proof span is lines 7447-7542, the complete author proof (94 body lines).

Editorial note. Navigation stubs: 15 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 54 of queue_spaceperfect.json (pb=4370, stmt 7421-7445, proof 7447-7542).

Editorial note. stacks-spaces-perfect wave sp01 (2026-10-14): fresh adjudication accept per lane-s3/sp01_judgments; pool per Chief 12:4x unfreeze (reserve-pool workflow, six-check clean); spaces-perfect.tex 126 with-proof rows, 70 b107-occupied labels avoided, 56 free rows judged 32 accept / 24 reject.

Editorial note. Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'proposition-detecting-bounded-below' also exists in stacks-perfect — a different proposition in a different chapter file.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 08GM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1513–1528. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0ATH: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7542–7542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GFG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7276–7284. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 09IX: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3877–3884. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 09MB: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4328–4338. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08GP: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 1563–1577. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08HK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3610–3616. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08GG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 2135–2145. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08FA: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 790–804. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GEK: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7542–7542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GEQ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7542–7542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0GFC: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 4141–4153. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 08JJ: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7542–7542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0B6E: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 7542–7542. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 09IY: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3891–3901. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. Let $G \in D_{perf}(\mathcal{O}_X)$ be a perfect complex which generates $D_{QCoh}(\mathcal{O}_X)$. Let $E \in D_{QCoh}(\mathcal{O}_X)$. The following are equivalent*

1. *$E \in D^+_{QCoh}(\mathcal{O}_X)$,*
2. *$\mathrm{Hom}_{D(\mathcal{O}_X)}(G[-i], E) = 0$ for $i \ll 0$,*
3. *$\mathrm{Ext}^i_X(G, E) = 0$ for $i \ll 0$,*
4. *$R\mathrm{Hom}_X(G, E)$ is in $D^+(\mathbf{Z})$,*
5. *$H^i(X, G^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E) = 0$ for $i \ll 0$,*
6. *$R\Gamma(X, G^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E)$ is in $D^+(\mathbf{Z})$,*
7. *for every perfect object P of $D(\mathcal{O}_X)$*
   (a) *the assertions (2), (3), (4) hold with G replaced by P, and*
   (b) *$H^i(X, P \otimes^{\mathbf{L}}_{\mathcal{O}_X} E) = 0$ for $i \ll 0$,*
   (c) *$R\Gamma(X, P \otimes^{\mathbf{L}}_{\mathcal{O}_X} E)$ is in $D^+(\mathbf{Z})$.*

Proof. Assume (1). Since $\mathrm{Hom}_{D(\mathcal{O}_X)}(G[-i], E) = \mathrm{Hom}_{D(\mathcal{O}_X)}(G, E[i])$ we see that this is zero for $i \ll 0$ by Lemma 09MB. This proves that (1) implies (2).

Parts (2), (3), (4) are equivalent by the discussion in Cohomology on Sites, Section 0B6E. Part (5) and (6) are equivalent as $H^i(X, -) = H^i(R\Gamma(X, -))$ by definition. The equivalent conditions (2), (3), (4) are equivalent to the equivalent conditions (5), (6) by Cohomology on Sites, Lemma 08JJ and the fact that $(G[-i])^\vee = G^\vee[i]$.

It is clear that (7) implies (2). Conversely, let us prove that the equivalent conditions (2) – (6) imply (7). Recall that G is a classical generator for $D_{perf}(\mathcal{O}_X)$ by Remark 0GFC. For $P \in D_{perf}(\mathcal{O}_X)$ let $T(P)$ be the assertion that $R\mathrm{Hom}_X(P, E)$ is in $D^+(\mathbf{Z})$. Clearly, T is inherited by direct sums, satisfies the 2-out-of-three property for distinguished triangles, is inherited by direct summands, and is preserved by shifts. Hence by Derived Categories, Remark 0ATH we see that (4) implies T holds on all of $D_{perf}(\mathcal{O}_X)$. The same argument works for all other properties, except that for property (7)(b) and (7)(c) we also use that $P \mapsto P^\vee$ is a self equivalence of $D_{perf}(\mathcal{O}_X)$. Small detail omitted.

We will prove the equivalent conditions (2) – (7) imply (1) using the induction principle of Lemma 08GP.

First, we prove (2) – (7) $\Rightarrow$ (1) if X is affine. This follows from the case of schemes, see Derived Categories of Schemes, Proposition 0GEQ.

Now assume $(U \subset X, j : V \to X)$ is an elementary distinguished square of quasi-compact and quasi-separated algebraic spaces over S and assume the implication (2) – (7) $\Rightarrow$ (1) is known for U, V, and $U \times_X V$. To finish the proof we have to show the implication (2) – (7) $\Rightarrow$ (1) for X. Suppose $E \in D_{QCoh}(\mathcal{O}_X)$ satisfies (2) – (7). By Lemma 09IX and Theorem 09IY there exists a perfect complex Q on X such that $Q|_U$ generates $D_{QCoh}(\mathcal{O}_U)$.

Say $V = \mathrm{Spec}(A)$. Let $Z \subset V$ be the reduced closed subscheme which is the inverse image of $X \setminus U$ and maps isomorphically to it (see Definition 08GM). This is a retrocompact closed subset of V. Choose $f_1, \ldots, f_r \in A$ such that $Z = V(f_1, \ldots, f_r)$. Let $K \in D(\mathcal{O}_V)$ be the perfect object corresponding to the Koszul complex on $f_1, \ldots, f_r$ over A. Note that since K is supported on Z, the pushforward $K' = Rj_*K$ is a perfect object of $D(\mathcal{O}_X)$ whose restriction to V is K (see Lemmas 08HK and 08GG). By assumption, we know $R\mathrm{Hom}_{\mathcal{O}_X}(Q, E)$ and $R\mathrm{Hom}_{\mathcal{O}_X}(K', E)$ are bounded below.

By Lemma 08GG we have $K' = j_!K$ and hence

$$\mathrm{Hom}_{D(\mathcal{O}_X)}(K'[-i], E) = \mathrm{Hom}_{D(\mathcal{O}_V)}(K[-i], E|_V) = 0$$

for $i \ll 0$. Therefore, we may apply Derived Categories of Schemes, Lemma 0GEK to $E|_V$ to obtain an integer a such that $\tau_{\leq a}(E|_V) = \tau_{\leq a}R(U \times_X V \to V)_*(E|_{U \times_X V})$. Then $\tau_{\leq a}E = \tau_{\leq a}R(U \to X)_*(E|_U)$ (check that the canonical map is an isomorphism after restricting to U and to V). Hence using Lemma 0GFG twice we see that

$$\mathrm{Hom}_{D(\mathcal{O}_U)}(Q|_U[-i], E|_U) = \mathrm{Hom}_{D(\mathcal{O}_X)}(Q[-i], R(U \to X)_*(E|_U)) = 0$$

for $i \ll 0$. Since the Proposition holds for U and the generator $Q|_U$, we have $E|_U \in D^+_{QCoh}(\mathcal{O}_U)$. But then since the functor $R(U \to X)_*$ preserves D^+_{QCoh} (by Lemma 08FA), we get $\tau_{\leq a}E \in D^+_{QCoh}(\mathcal{O}_X)$. Thus $E \in D^+_{QCoh}(\mathcal{O}_X)$. $\square$